

Ajeet Krishnasamy

Dallas, TX | +1-647-471-2271 | ajeetkrish@icloud.com | [LinkedIn](#) | [Portfolio](#)

Mechanical engineering graduate specializing in CFD, FEA, and autonomous flight dynamics, with hands-on experience developing high-performance C++ physics solvers, Python simulation tools, and control systems.

EXPERIENCE

Systems Engineer, Medical Equipment (Intern)

Oct 2020 – Sept 2021

CenTrak

Newtown, PA, United States

- Engineered automated reliability analysis pipelines for medical device telemetry, processing real-time performance data from 10,000+ RTLS tags to quantify failure rates and improve system monitoring precision.
- Developed automated performance dashboards for the Connect Pulse™ RTLS platform, enabling engineering teams to identify device degradation trends and reduce unplanned maintenance events.
- Conducted battery discharge and device health analysis across 5,000+ medical-grade IoT devices, identifying failure precursors and enabling predictive replacement scheduling that prevented equipment outages.
- Standardized structured query and analysis workflows for multi-terabyte RTLS datasets, reducing troubleshooting cycle time through automated anomaly detection and root-cause analysis scripts.

Systems Engineer, Medical Equipment (Intern)

July 2019 – Sept 2019

CenTrak

Newtown, PA, United States

- Performed system-level characterization of RTLS localization algorithms using statistical regression analysis, identifying coverage gaps and refining location accuracy through receiver parameter tuning.
- Designed and executed validation test protocols for RTLS hardware deployment, correlating laboratory calibration data with field performance metrics to close the lab-to-field validation gap.
- Synthesized system performance data into technical recommendations for hardware upgrades and feature rollouts, supporting engineering decisions with quantitative evidence.

PROJECTS

AK-Vortex: High-Performance C++ Lattice Boltzmann CFD Suite | C++, Python, CFD

- Built a D2Q9 Lattice Boltzmann CFD solver in C++20 with MRT collision, curved boundary handling, Smagorinsky LES turbulence modeling, and adaptive mesh refinement; validated against 14 canonical benchmarks (Blasius, Ghia, Armaly, Tritton).
- Developed a PINN surrogate model achieving 300–600x real-time speedup over the solver for design-space exploration, and a cross-platform desktop application with interactive geometry editing and parameter sweeps.
- 219 Google Test cases with CI; 12-page interactive portfolio with flow visualization, validation tables, and theoretical documentation.

Crucible-FEA: 2D/3D Finite Element Structural Solver | C++, Python, FEA

- Developed a C++20 finite element solver supporting 6 element types (bar, quad, serendipity, triangle, hex, tet) with sparse matrix assembly, direct and iterative solvers, and multiple preconditioners.
- Implemented adaptive mesh refinement, geometric nonlinearity, transient dynamics, modal analysis, and contact mechanics; validated against 10 analytical benchmarks within 1%.
- Built a Qt 6 desktop application with mesh editor, boundary condition assignment, real-time stress/displacement contour visualization, and analysis plots.

Zenith: Orbital Mechanics & Flight Dynamics Engine | C++20, Python

- Built an orbital propagation pipeline in C++20 covering SGP4/SDP4, numerical integrators (RK4, RK45), and perturbation models (J2, drag, SRP, third-body); validated against Vallado Spacetrack Report #3.
- Developed analysis modules for conjunction assessment, maneuver planning (Hohmann, Lambert), Monte Carlo uncertainty propagation, ground tracks, and constellation coverage analysis.
- Built an interactive 3D orbit viewer with real-time propagation, force model toggles, and preset satellite database; 139 Google Test cases.

TECHNICAL SKILLS

Aerodynamics & CFD: ANSYS CFD, STAR-CCM+, SU2, OpenFOAM, Lattice Boltzmann, Gmsh, ParaView

Programming & CAD/FEA: Python, C++, MATLAB, SolidWorks, Autodesk Fusion, Git, CMake

Dynamics & Control: LQR Optimal Control, State-Space Dynamics, APF Guidance, Graph Laplacian, Monte Carlo

Manufacturing: CNC Machining, 3D Printing, Lathe & Mill Operations, MIG Welding

EDUCATION

University of Ottawa

Ottawa, ON, Canada

Bachelor of Applied Science (BASc), Biomedical Mechanical Engineering

May 2026

Coursework: Fluid Mechanics, Heat Transfer, Control Systems, Vibration Analysis, Machine Design, CAD/FEA

Accreditation: CEAB-accredited; ABET-equivalent